TIPA Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Tailored Cell-Based Assays and unique in vitro cell culture tool to support Korean biomedical innovation and medical device development			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio ·Medical		Bio ·Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Bio - Biomaterials	Bio-processes / Bio-Equipment	Other Bio-process / Technologies
Concept of Proposed Technology	The proposed technology is a modular, cell-based in vitro testing platform combined with a unique in vitro cell culture tool, designed to support the development and evaluation of biomaterials, medical devices, and drug-eluting systems. The platform enables tailored, application-specific cell-based assays with a strong focus on biological relevance, reproducibility, and quantitative comparability. It allows direct investigation of material–cell interactions, immune responses, and biocompatibility aspects, as well as the biological effects of released active substances, such as small molecules or bioactive compounds, under standardized and scalable test conditions. By integrating customizable assay strategies with a dedicated in vitro cell culture tool, the technology bridges the gap between early-stage R&D and application-oriented preclinical evaluation. It supports			

	the assessment of both passive and active biomedical products, including implantable and non-implantable medical devices, surface-modified materials, wound care products, and drug-eluting systems. The platform is designed to be adaptable to different biomedical applications and product classes. It enables Korean SMEs to generate robust, decision-relevant in vitro data for material selection (suitable candidates / formulations), device optimization, evaluation of drug–material combinations, and early risk reduction in biomedical product development.
Rationale for Joint R&D Project	Korean biomedical SMEs increasingly require reliable, biologically relevant in vitro testing strategies to support innovation, shorten development cycles, and reduce development risks for new medical devices and biomaterials. However, many companies face challenges in implementing standardized, reproducible, and application-specific cell-based assays within their own R&D environments. This joint R&D project aims to combine Fraunhofer's expertise in cell-based in vitro testing and assay standardization with the application-driven development activities of Korean SMEs. Through close collaboration, the platform can be adapted to specific product classes, regulatory requirements, and industrial workflows. The joint development approach enables knowledge transfer, validation under industrially relevant conditions, and co-creation of testing solutions that are directly applicable within Korean biomedical R&D and product development pipelines.
Key Words	1. Development of modular in vitro testing strategies 2. Bio- and Immune-compatibility studies 3. Standardized cell-based testing (tool) 4. Reproducible quantitative data 5. Comparative evaluation of biomaterials

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	The overall objective of this joint R&D project is to further develop and transfer a modular, cell-based in vitro testing platform and an associated in vitro cell culture tool from laboratory validation to an industry-ready, application-oriented testing solution for biomedical R&D. Specific objectives include: 1. Adaptation and customization of cell-based assay modules for different biomedical applications, including biomaterials, medical devices, and drug-eluting systems. 2. Integration of the in vitro cell culture tool into standardized and scalable testing workflows, enabling reproducible assay implementation. 3. Generation of robust, quantitative, and comparable in vitro data addressing biocompatibility, immune response, and biological effects of released active substances, such as small molecules. 4. Demonstration of modular in vitro test strategies within collaborative R&D projects, including adapted laboratory components, in vitro cell culture tools, and customized test scenarios, while ensuring controlled and protected access to proprietary methodologies. 5. Design and validation of customized cell-based assay strategies addressing material-specific and drug-related safety and functionality questions, including direct cell–material and cell–drug interaction studies, ensuring scientific and regulatory robustness. 6. Preparation for industry-relevant implementation of selected cell-based assay modules, including considerations for scalability, usability, and adaptation to SME-specific R&D laboratory environments, while ensuring appropriate protection of proprietary methodologies.			
R&D Target Specifications	The R&D project aims to achieve the following target specifications: • Establishment of standardized and reproducible in vitro test scenarios addressing bio- and immunological questions based on direct cell–material and cell–drug interactions. • Development of modular, application-adaptable in vitro test systems suitable for the evaluation of biomaterials, medical devices, and drug-based or drug-eluting products. • Generation of reliable, quantitative, and comparable datasets, including raw data, suitable for internal R&D decision-making and technology development within industrial environments. • Demonstration of the modular test strategy and adapted analytical system components within collaborative R&D projects under real development conditions. • Controlled implementation of tailored test strategies aligned with agreed collaboration objectives, ensuring appropriate protection of proprietary methodologies while enabling effective joint R&D. • Provision of validated, biologically relevant in vitro testing outputs that support informed material selection, product optimization, and early-stage risk assessment in biomedical product development, such as functional readouts related to cytotoxicity and metabolic activity, cell adhesion and proliferation on material surfaces, and immune cell activation reflected by cytokine release profiles.			

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R&D Plan	The R&D activities will be structured into two consecutive phases in accordance with the TIPA program framework. Phase I – Pre-study (6 months) • Definition and prioritization of target application areas in collaboration with Korean SMEs. • Selection and adaptation of representative cell-based assay modules addressing biocompatibility, immune response, and other biologically relevant effects. • Technical optimization of the in vitro cell culture tool for application-specific testing scenarios. • Establishment of standardized experimental workflows and quality criteria for reproducibility and data comparability. • Evaluation of technical feasibility, scalability, and industrial relevance as a basis for Phase II application. Phase II – Main R&D (30 months) • Further development and refinement of the modular cell-based assay platform and associated in vitro cell culture tool based on the outcomes of Phase I. • Customization of assay modules and test workflows for different biomedical product classes, including biomaterials, medical devices, and drug-eluting systems, in close collaboration with Korean SMEs. • Demonstration and validation of the modular assay platform within collaborative R&D projects under industry-relevant conditions. • Exemplary application scenarios may include the development and evaluation of medical devices such as stents, addressing material-specific modifications, degradation behavior, and biological responses using tailored cell-based in vitro testing strategies in real-world industrial development contexts. • Preparation and demonstration of industry-relevant implementation of selected cell-based assay modules, including adaptation of test scenarios and workflows to SME-specific R&D laboratory environments, enabling potential post-project integration of validated assays into industrial R&D settings. • Validation of platform robustness, usability, and reproducibility under application-oriented conditions, reaching TRL 7.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer institute will take the lead in the scientific and technical development, optimization, and validation of modular cell-based in vitro test systems and associated in vitro cell culture tools for biomaterial, medical device, and drug-related applications. Their role will focus on adapting test system components, assay modules, and experimental workflows to the specific needs of Korean SMEs and their products, while ensuring scientific robustness, reproducibility, and applicability under industry-relevant conditions. Fraunhofer will establish standardized in vitro test scenarios addressing bio- and immunological questions based on direct cell–material and cell–drug interactions and will conduct targeted data acquisition within collaborative R&D projects. In close collaboration with Korean SMEs, Fraunhofer will execute demonstration projects and support the preparation of industry-relevant assay implementation, enabling potential post-project integration of selected assay modules into SME R&D laboratory environments while ensuring controlled access to proprietary methodologies.
Expected R&R of Korean SME Partner	Korean SMEs will contribute application-specific requirements, product concepts, materials, devices, or formulations to be evaluated within the collaborative R&D framework. They will support the adaptation of test strategies to real-world industrial development scenarios and participate in validation activities under application-oriented conditions. The SMEs will collaborate closely in defining relevant use cases, assessing industrial feasibility, and evaluating the applicability of the developed testing platform for their product development pipelines.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to result in a validated, modular cell-based in vitro testing platform and associated in vitro cell culture tool that can be adapted to a wide range of biomedical applications, including biomaterials, medical devices, and drug-eluting systems. Key technological outcomes include standardized and customizable assay modules, reproducible testing workflows, and demonstrator case studies illustrating the applicability of the platform under industry-relevant conditions. The project will enable the generation of reliable, decision-relevant in vitro data supporting material selection, product optimization, and early-stage risk assessment. From an industrial and economic perspective, the project supports Korean SMEs in accelerating biomedical product development, reducing development risks, and improving R&D efficiency. By enabling application-oriented in vitro testing strategies and preparing the implementation of selected assay modules within SME R&D laboratory environments, the project contributes to shortening development timelines and increasing the likelihood of successful market entry of biomedical products. While Fraunhofer does not commercialize a standalone product within this project, the developed platform and methodologies create clear commercialization pathways for Korean SMEs. These include improved development and validation of medical devices and biomaterials, enhanced competitiveness through robust preclinical data, and potential follow-up activities such as contract research, technology services, or longer-term collaboration models. Overall, the project strengthens international industrial collaboration between Fraunhofer and Korean SMEs and contributes to the advancement and competitiveness of the Korean biomedical and medical device industry through advanced, biologically relevant in vitro testing solutions.

6. References

References available upon request.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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